EDGE INTELLIGENCE LAB

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Log Data Analytics

Objective

Enable real-time log analysis at the edge to reduce latency, bandwidth usage, and improve faster decision-making for distributed systems (IoT, cloud-native apps, industrial systems).

Key Goals

- Process logs locally at edge devices
- Detect anomalies in real-time
- Reduce data transmission to cloud
- Enable predictive maintenance & alerts
- Ensure data privacy & security

Architecture Overview

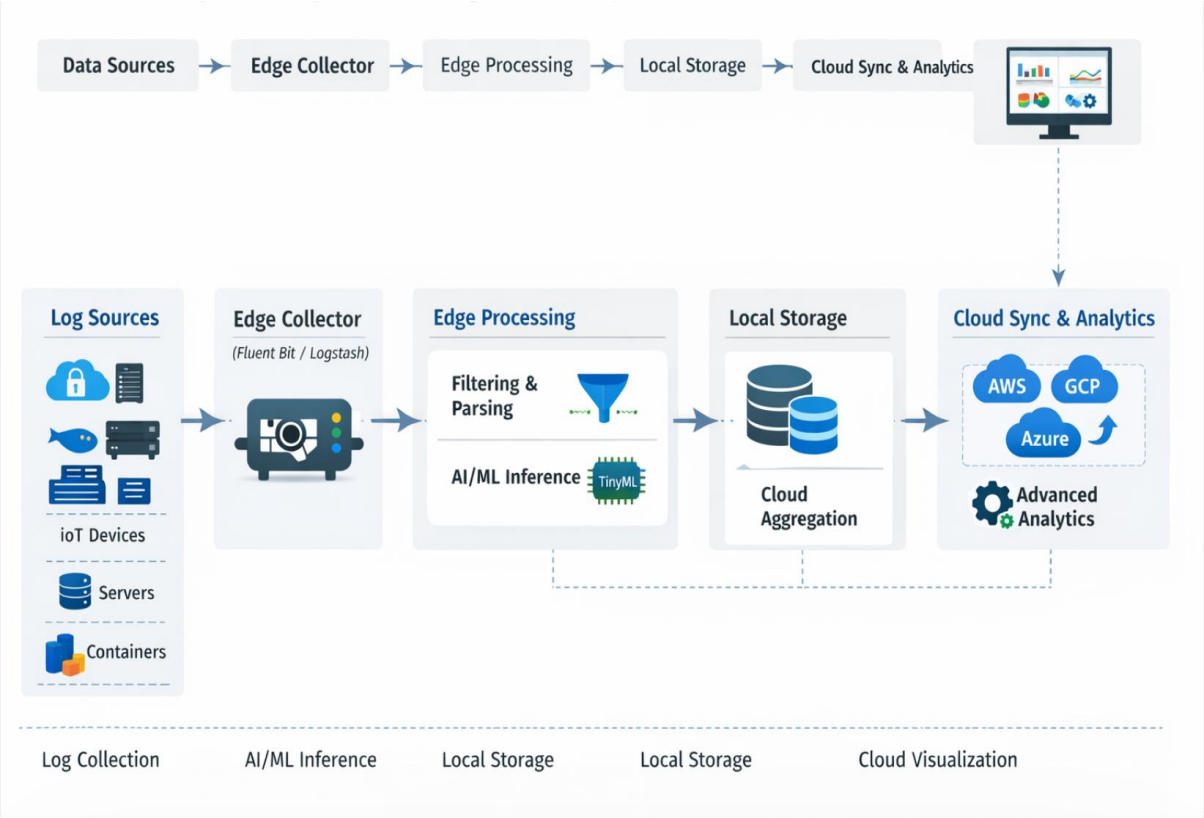
Flow:

Data Sources → Edge Collector → Edge Processing → Local Storage → Cloud Sync → Dashboard

Components:

- Log Sources: IoT devices, servers, containers
- Edge Collector: Fluent Bit / Logstash (lightweight)
- Edge Processing:
    - Filtering & parsing
    - AI/ML inference (TinyML / lightweight models)
- Local Storage: SQLite / Edge DB

- Cloud Layer:
    - Aggregation (AWS, GCP, Azure)
    - Advanced analytics
- Visualization: Grafana / Kibana dashboards



Technology Stack

- Edge Devices: Raspberry Pi, NVIDIA Jetson
- Log Collection: Fluent Bit, Filebeat
- Processing: Python, Node.js
- ML Models:
    - Anomaly detection (Isolation Forest, LSTM)
- Containers: Docker, K3s (Lightweight Kubernetes)
- Cloud Platforms:
    - AWS IoT / GCP Edge / Azure IoT Hub

Workflow

1. Logs generated from applications/devices
2. Edge agent collects logs
3. Preprocessing:
    - Remove noise
    - Format logs
4. Local analytics:
    - Pattern detection
    - Anomaly detection
5. Alerts triggered locally (if needed)
6. Only summarized/critical logs sent to cloud
7. Cloud performs deeper analytics & visualization

Use Cases

- Smart Manufacturing: Detect machine failures early
- Smart Cities: Traffic/log monitoring
- Healthcare IoT: Real-time patient device logs
- Cloud-native apps: Kubernetes pod log monitoring

Benefits

- Low latency (real-time insights)
- Reduced cloud costs
- Improved data privacy
- Lower bandwidth usage
- Scalable distributed analytics

Future Enhancements

- Federated learning at edge
- AI-driven root cause analysis
- Integration with AIOps platforms
- Self-healing systems

Success Metrics

- Reduction in cloud data transfer (%)
- Detection latency (ms)
- Accuracy of anomaly detection
- System uptime improvement